

[FIXTURE] Synthetic capture for Quorum's offline demo — not a real fetch.

Northwind Chat — Inference Cost Report, Q1 2026

Method

We instrumented 41 production deployments of assistant-style chat products between October 2025 and February 2026, recording per-request token accounting from provider usage objects rather than from client-side estimates. Deployments self-selected into the study; we make no claim that they represent the market.

Headline result

Across the 41 deployments, median spend on input tokens fell 52% after a cache breakpoint was introduced above the system prompt. The interquartile range is wide: the 25th percentile saved 31% and the 75th percentile saved 68%.

Breakeven

A cached prefix pays for itself once it is read roughly 5 times per write. Below that the $1.25\times$ write premium is never recovered, and a product that rewrites its prefix on every request pays about 25% more than it would have paid with no caching at all.

Sensitivity

The saving is dominated by the share of the prompt that is static. Deployments whose static prefix was under 30% of median prompt length saved less than 15%, whatever their hit rate.

The low-hit-rate regime

Below a 40% cache hit rate the median deployment saved nothing measurable: write premiums and read discounts cancelled within the noise of week-to-week traffic. Nine of the 41 deployments spent more with caching enabled than they had without it.

Bursty traffic

Hit rate tracks session density, not request volume. A product with the same daily volume spread thinly across time zones sat at a 22% hit rate; the same volume concentrated into working hours sat at 71%.

Limitations

Self-selection is the obvious one. Deployments that switched caching on and saw nothing had no incentive to enrol, so the headline figure is an upper bound rather than an expectation.